Section 18.5–18.6 Rewrite Using Senior Design Report Notation

April 21, 2026

Scope

This document rewrites only Section 18.5 and Section 18.6 using the symbol conventions that appear in `Senior Design Project Report Draft 1 - Cited.docx`, especially the bed/branch heat relations and the two-node matrix notation.

Notation Mapping Used in This Rewrite

Section 18 symbol (current physics docs)	Rewritten symbol (Senior Design draft style)
$Q_{\text{to bed},h}(d)$, $Q_{\text{to bed},c}(d)$	$Q^{(h)}_{\text{bed,total}}(d)$, $Q^{(c)}_{\text{bed,total}}(d)$
$Q_{\text{wall}}, Q_{\text{jet}}, Q_{\text{evap}}$ (segment terms)	$q_{\text{wall},i}, q_{\text{jet},i}, q_{\text{evap},i}$
$Q_{\text{bed}}(d)$ (applied net load)	$Q_{\text{bed,total,applied}}(d)$
$T_b(d), T_t(d)$	$T_{\text{bed},b}(d), T_{\text{bed},t}(d)$
$UA_{b,\infty}, UA_{t,\infty}, UA_{bt}$	$(UA)_{b,\infty}, (UA)_{t,\infty}, (UA)_{12}$
$P_h(d)$	$P_{\text{tot},h}(d)$
$P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d)$	$P_{\text{tot},c}(d)$

18.5 Capacity Re-solve and Duty Application (Docx Notation)

Equation-number convention. When a formula comes directly from the Senior Design draft notation, I keep that draft equation number in the tag. When the formula is Section 18-specific glue logic (duty, clipping, aggregation), I assign a local tag 18.5∗ or 18.6∗.

Step 18.5-A: Required daily heating/cooling load. Using the same daily ambient $T_\infty(d)$, define required loads:

$$Q_{\text{load},h}(d) = \max((UA)_{\text{tot},*}\,(T_{h,\text{set}} - T_\infty(d))\,, 0)\,, \tag{18.5A-1}$$

$$Q_{\text{load},c}(d) = \max((UA)_{\text{tot},*}\,(T_\infty(d) - T_{c,\text{set}})\,, 0)\,. \tag{18.5A-2}$$

Here $(UA)_{\text{tot},*}$ is the same bin-loss conductance term used in the report's $Q_{\text{loss,bin}}$ relation.

Step 18.5-B: Re-solved bed-transfer capacity at day ambient. At each day d, solve branch/segment transfer using the report's segment equations:

$$\dot{m}_{\text{mean},i}(d) = \frac{\dot{m}_{\text{rem},i}(d) + \dot{m}_{\text{rem},i+1}(d)}{2}, \tag{47}$$

$$\Theta_i(d) = \exp\left[-\frac{U_{\text{aer}}\Pi_{\text{out}}\Delta x_{\text{aer}}}{\dot{m}_{\text{mean},i}(d)C_P}\right], \tag{49}$$

$$T_{\text{air,post wall},i}(d) = T_{\text{bed,ref}}(d) + \left(T_{\text{air},i}(d) - T_{\text{bed,ref}}(d)\right)\Theta_i(d), \tag{50}$$

$$T_{\text{rel},i}(d) = \frac{T_{\text{air},i}(d) + T_{\text{air,post wall},i}(d)}{2}, \tag{52}$$

$$q_{\text{wall},i}(d) = \dot{m}_{\text{mean},i}(d)C_P\left(T_{\text{air},i}(d) - T_{\text{air,post wall},i}(d)\right), \tag{44}$$

$$q_{\text{jet},i}(d) = \dot{m}_{\text{rel},i}(d)C_P\left(T_{\text{rel},i}(d) - T_{\text{bed,ref}}(d)\right), \tag{43}$$

$$q_{\text{bed},i}(d) = q_{\text{wall},i}(d) + q_{\text{jet},i}(d) - q_{\text{evap},i}(d), \tag{42}$$

$$Q_{\text{bed,branch}}(d) = \sum_{i=1}^{N_{\text{aer}}} q_{\text{bed},i}(d), \tag{45}$$

$$Q_{\text{bed,total}}(d) = N_{\text{tubes}}\, Q_{\text{bed,branch}}(d). \tag{54}$$

Use mode-specific re-solves:

$$Q^{(h)}_{\text{bed,total}}(d), \qquad Q^{(c)}_{\text{bed,total}}(d), \tag{18.5B-1}$$

and convert to positive capacities:

$$Q_{h,\text{cap}}(d) = \max\left(Q^{(h)}_{\text{bed,total}}(d), 0\right), \qquad Q_{c,\text{cap}}(d) = \max\left(-Q^{(c)}_{\text{bed,total}}(d), 0\right). \tag{18.5B-2}$$

Step 18.5-C: Re-solved electrical power at day ambient. Heating coil power uses the report's electrical sum style:

$$P_{h,\text{coil}}(d) = \sum_g n_{\text{coil},g}\, P_{\text{coil},g}(d), \tag{31}$$

For blower $x \in \{\text{hb}, \text{ab}\}$ (hb: heating blower, ab: cooling assist blower), the same operating-flow and power model is used:

$$\Delta p_{\text{avail},x}(\dot{V}) = \max\left(\Delta p_{\text{shutoff},x}\left(1 - \frac{\dot{V}}{\dot{V}_{\text{free},x}}\right), 0\right), \tag{18.5C-1a}$$

$$\dot{V}_{\text{free},x} = \begin{cases} \dot{V}_{\text{rated},x}\,\dfrac{\Delta p_{\text{shutoff},x}}{\Delta p_{\text{shutoff},x} - \Delta p_{\text{rated},x}}, & \Delta p_{\text{shutoff},x} > \Delta p_{\text{rated},x}, \\ \dot{V}_{\text{rated},x}, & \text{otherwise}, \end{cases} \tag{18.5C-1a2}$$

Eq. (18.5C-1a2) comes from a two-point linear fan-curve fit:

$$(\dot{V}, \Delta p) = (0, \Delta p_{\text{shutoff},x}), \qquad (\dot{V}, \Delta p) = (\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x}).$$

The slope is

$$m_x = \frac{\Delta p_{\text{rated},x} - \Delta p_{\text{shutoff},x}}{\dot{V}_{\text{rated},x}}, \qquad \Delta p_x(\dot{V}) = \Delta p_{\text{shutoff},x} + m_x\dot{V}.$$

Setting $\Delta p_x(\dot{V}_{\text{free},x}) = 0$ gives Eq. (18.5C-1a2). If $\Delta p_{\text{shutoff},x} \le \Delta p_{\text{rated},x}$, this extrapolation is not physically reliable, so implementation falls back to $\dot{V}_{\text{free},x} = \dot{V}_{\text{rated},x}$. Conceptually: $\Delta p_{\text{shutoff},x}$ and $\Delta p_{\text{rated},x}$ are two static-pressure-rise points (Pa), not absolute pressures. They are read from one blower curve at $\dot{V} = 0$ and $\dot{V} = \dot{V}_{\text{rated},x}$, and those two points define the linear pressure-flow relation used in Eqs. (18.5C-1a) and (18.5C-1a2). Implementation sequence: when datasheet $\dot{V}_{\text{free},x}$ exists, the blower line is used directly from $(0, \Delta p_{\text{shutoff},x})$ to $(\dot{V}_{\text{free},x}, 0)$. When $\dot{V}_{\text{free},x}$ is missing,

Eq. (18.5C-1a2) infers it from $(\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x})$ and $\Delta p_{\text{shutoff},x}$, then Eq. (18.5C-1a) uses that inferred endpoint. when explicit free-air flow is not provided.

$$R_x(\dot{V}) = \Delta p_{\text{sys},x}(\dot{V}) - \Delta p_{\text{avail},x}(\dot{V}), \tag{18.5C-1b}$$

$$\Delta p_{\text{dist},c} = \Delta p_{\text{header}} + \Delta p_{\text{header}\to\text{splitter}} + \Delta p_{\text{splitter body}} + \Delta p_{\text{connector fric}} + \Delta p_{\text{connector}\to\text{branch}}, \tag{56}$$

$$p_{\text{in},c} = \Delta p_{\text{aer,reported}}, \tag{18.5C-1b0}$$

$$\Delta p_{\text{sys,ab}}(\dot{V}) = \Delta p_{\text{tot,cool}}(\dot{V}) = \Delta p_{\text{dist},c}(\dot{V}) + p_{\text{in},c}(\dot{V}), \tag{76}$$

$$\dot{m}_{\text{rel},s} = C_d A_{h,s} \sqrt{2\rho_s \max(p_s - p_{\text{bed},s}, 0)}, \tag{15}$$

$$\frac{\Delta p_{\text{Ergun},s}}{L_s} = \frac{150(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu_s u_s}{d_p^2} + \frac{1.75(1-\varepsilon)}{\varepsilon^3}\frac{\rho_s u_s^2}{d_p}, \qquad u_s = \frac{\dot{m}_{\text{rel},s}}{\rho_s A_{\text{trib},s}}, \tag{62}$$

$$p_{\text{bed},s} = p_{\text{out,bnd}} + \Delta p_{\text{Ergun},s}, \tag{60}$$

$$\Delta p_{\text{path},g} = \Delta p_{\text{dist,private},g} + \Delta p_{\text{heater,tube},g} + \Delta p_{\text{heater,coil},g} + \Delta p_{\text{heater}\to\text{aer},g} + p_{\text{in,aer},g}, \tag{80}$$

$$\Delta p_{\text{sys,hb}}(\dot{V}) = \Delta p_{\text{tot,heat}}(\dot{V}) = \Delta p_{\text{dist,common}}(\dot{V}) + \Delta p_{\text{path,ctrl}}(\dot{V}), \tag{81}$$

In the closed-end shooting step (described in the source text directly after Eq. (10)), the inlet pressure is solved so terminal remaining flow meets the configured mass-balance tolerance. In the branch march implementation, $\dot{m}_{\text{rel},s}$ is bounded by the segment-remaining flow $\dot{m}_{\text{rem},s}$, and $\Delta p_{\text{path,ctrl}}$ is the largest active-group value of $\Delta p_{\text{path},g}$ used for the parallel-network pressure closure.

$$\mathcal{F}_x(d) = \left\{ \dot{V} \in [0, \dot{V}_{\text{req},x}(d)] \ : \ R_x(\dot{V}) \le 0 \right\}, \tag{18.5C-1c0}$$

$$\dot{V}_{\text{feas},x}(d) = \begin{cases} \max \mathcal{F}_x(d), & \mathcal{F}_x(d) \neq \varnothing, \\ 0, & \mathcal{F}_x(d) = \varnothing, \end{cases} \tag{18.5C-1c1}$$

$$\dot{V}_{\text{ach},x}(d) = \begin{cases} \dot{V}_{\text{req},x}(d), & \text{if blower } x \text{ is disabled, fan data are unavailable, or } R_x(\dot{V}_{\text{req},x}(d)) \le 0, \\ \dot{V}_{\text{feas},x}(d), & \text{otherwise,} \end{cases} \tag{18.5C-1c}$$

$$\Delta p_{\text{op},x}(d) = \begin{cases} 0, & \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) \le 0, \\ \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)), & 0 < \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) < \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)), \\ \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)), & \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) \ge \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)). \end{cases} \tag{18.5C-1d}$$

$$P_{\text{shaft},x}(d) = \dot{V}_{\text{ach},x}(d)\, \Delta p_{\text{op},x}(d), \tag{18.5C-1e}$$

$$P_{x,\text{elec}}(d) = \min\left(P_{\text{rated},x},\ P_{\text{idle},x} + \frac{P_{\text{shaft},x}(d)}{\eta_x} \right), \tag{18.5C-1f}$$

$$P_{x,\text{elec}}(d) = \min\left(P_{\text{rated},x}, P_{\text{idle},x} + (P_{\text{rated},x} - P_{\text{idle},x}) \min\left(\max\left(\frac{\dot{V}_{\text{ach},x}(d)}{\dot{V}_{\text{free},x}}, 0 \right), 1 \right) \right), \tag{18.5C-1g}$$

Eq. (18.5C-1g) is the code fallback used only when η_x is unavailable.

$$P_{x,\mathrm{elec}}(d) = \begin{cases} 0, & \text{if blower } x \text{ is disabled,} \\ & \text{or } \dot{V}_{\mathrm{ach},x}(d) \le 0 \\ \min\left(P_{\mathrm{rated},x},\, P_{\mathrm{idle},x} + \dfrac{P_{\mathrm{shaft},x}(d)}{\eta_x}\right), & \text{if } \eta_x \text{ is provided,} \\ \min\left(P_{\mathrm{rated},x}, P_{\mathrm{idle},x} + (P_{\mathrm{rated},x} - P_{\mathrm{idle},x}) \min\left(\max\left(\dfrac{\dot{V}_{\mathrm{ach},x}(d)}{\dot{V}_{\mathrm{free},x}}, 0\right), 1\right)\right), & \text{otherwise.} \end{cases} \tag{18.5C-1h}$$

This piecewise form is the exact implementation logic. $P_{\mathrm{idle},x}$ is *not* consumed when the blower is off or solved flow is zero, because the implemented return is 0 in that case. Conceptually, $P_{\mathrm{idle},x}$ is the baseline non-hydraulic electrical draw when the blower is running at very low flow (motor/controller and parasitic losses before pressure-flow work). In configuration this is `idlePower_W`; code enforces $P_{\mathrm{idle},x} \ge 0$ and always caps final $P_{x,\mathrm{elec}}(d)$ at $P_{\mathrm{rated},x}$. For physically consistent calibration, select $P_{\mathrm{idle},x}$ from a measured low-flow operating point and keep $0 \le P_{\mathrm{idle},x} \le P_{\mathrm{rated},x}$. Thus, total heating electrical power is

$$P_{\mathrm{tot},h}(d) = P_{h,\mathrm{coil}}(d) + P_{\mathrm{hb,elec}}(d). \tag{18.5C-2}$$

Cooling electrical terms are:

$$\dot{V}_{\mathrm{rated}} = 0.00047194745\, V_{\mathrm{rated,CFM}}, \qquad \dot{m}_{\mathrm{rated}}(d) = \rho_{\mathrm{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\mathrm{rated}}. \tag{18.5C-3a0}$$

$$\dot{Q}_{\mathrm{rated,sens}}(d) = \max\Big(\dot{Q}_{\mathrm{rated,total}} - \dot{Q}_{\mathrm{rated,lat}}(d),\, 0\Big). \tag{18.5C-3a0a}$$

$$T_{\mathrm{target,derived}}(d) = T_\infty(d) - \frac{\dot{Q}_{\mathrm{rated,sens}}(d)}{\dot{m}_{\mathrm{rated}}(d) c_p(d)}, \quad \dot{m}_{\mathrm{rated}}(d) c_p(d) > 0. \tag{18.5C-3a0b}$$

$$T_{\mathrm{target}}(d) = \begin{cases} T_{\mathrm{target,derived}}(d), & \text{if } \chi_{\mathrm{derive}} = 1 \text{ and } T_{\mathrm{target,derived}}(d) \text{ is finite,} \\ T_{\mathrm{target,cfg}}, & \text{if a configured target is provided,} \\ T_\infty(d), & \text{otherwise.} \end{cases} \tag{18.5C-3a0c}$$

Eq. (18.5C-3a0c) matches code precedence: derived target overrides configured target only when `deriveTargetFromRatedSpecs` is enabled and derivation is finite.

$$\dot{m}(d) = \rho_{\mathrm{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\mathrm{ach,ab}}(d), \qquad \dot{m}_{\mathrm{dry}}(d) = \frac{\dot{m}(d)}{1 + w_{\mathrm{in}}(d)}. \tag{18.5C-3a0d}$$

$\dot{V}_{\mathrm{ach,ab}}(d)$ is the achieved assist-blower flow from the residual closure in Eqs. (18.5C-1b)–(18.5C-1c3) over $[0, \dot{V}_{\mathrm{req,ab}}(d)]$.

$$\dot{Q}_{\mathrm{spot,req}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\mathrm{target}}(d), 0), \tag{18.5C-3a}$$

$$\dot{Q}_{\mathrm{spot,act}}(d) = \min\Big(\dot{Q}_{\mathrm{spot,req}}(d), \dot{Q}_{\mathrm{spot,max}}(d)\Big), \tag{18.5C-3b}$$

$$\dot{Q}_{\mathrm{spot,max}}(d) = \begin{cases} Q_{\mathrm{spot,max,cfg}}, & \text{if } \texttt{maxCoolingCapacity_W} \text{ is provided,} \\ \max\Big(\dot{Q}_{\mathrm{rated,total}} - \dot{Q}_{\mathrm{rated,lat}}(d),\, 0\Big), & \text{if rated-spec path is used,} \\ +\infty, & \text{otherwise,} \end{cases} \tag{18.5C-3b1}$$

$$\dot{Q}_{\text{rated,total}} = 0.29307107\, C_{\text{rated,BTU/h}},$$
$$\dot{m}_{\text{cond,rated}} = \frac{0.473176473\, D_{\text{rated,pint/day}}}{86400}, \tag{18.5C-3b2}$$
$$\dot{Q}_{\text{rated,lat}}(d) = \dot{m}_{\text{cond,rated}}\, h_{fg}(T_\infty(d)).$$

$$P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d). \tag{18.5C-3c00}$$

$$\dot{Q}_{\text{sensible}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{supply,actual}}(d), 0) = \dot{Q}_{\text{spot,act}}(d),$$
$$\text{COP} = \frac{\dot{Q}_{\text{cool}}}{P_{\text{in}}(d)},$$
$$\text{COP}_{\text{sensible}} = \frac{\dot{Q}_{\text{sensible}}(d)}{P_{\text{in}}(d)}, \tag{18.5C-3c0}$$
$$\text{COP} = \frac{\text{EER}}{3.412}.$$

In Eq. (18.5C-3c), COP is the configuration input and should be interpreted as sensible COP because $\dot{Q}_{\text{spot,act}}(d)$ is a sensible load.

$$P_{\text{spot cooler}}(d) = \frac{\dot{Q}_{\text{spot,act}}(d)}{\text{COP}}, \tag{18.5C-3c}$$

$$P_{\text{assist blower}}(d) = P_{\text{ab,elec}}(d), \tag{18.5C-3d}$$

$$P_{\text{tot},c}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d). \tag{18.5C-4}$$

Step 18.5-C definitions (all new symbols). $x \in \{\text{hb}, \text{ab}\}$: blower index. hb is heating blower, ab is cooling assist blower. $\dot{V}$ is volumetric flow [m^3/s]. Δp is pressure rise/drop [Pa]. $R_x(\dot{V})$ in Eq. (18.5C-1b) is blower-flow residual (system-required minus blower-available pressure). $\dot{V}_{\text{req},x}(d)$ is requested flow. $\mathcal{F}_x(d)$ in Eq. (18.5C-1c0) is the feasible-flow set. $\dot{V}_{\text{feas},x}(d)$ in Eq. (18.5C-1c1) is the maximum feasible-flow helper. $\dot{V}_{\text{ach},x}(d)$ is achieved flow used in the final day model.

$\Delta p_{\text{sys,ab}}$ is tied to the original pressure-balance numbering through Eqs. (56) and (76), plus the report-to-code mapping in Eq. (18.5C-1b0): Eq. (56) defines $\Delta p_{\text{dist},c}$, Eq. (18.5C-1b0) maps the reported aeration inlet pressure to $p_{\text{in},c}$, and Eq. (76) gives the cooling total $\Delta p_{\text{tot,cool}} = \Delta p_{\text{dist},c} + p_{\text{in},c}$. $p_{\text{out,bnd}}$ is the bed outlet-boundary pressure (open-top/vent/offset closure). $p_{\text{bed},s}$ is local bed-side pressure in segment s from Eq. (60). $\Delta p_{\text{Ergun},s}$ is local porous-bed backpressure from Eq. (62). L_s, ε, d_p, $A_{\text{trib},s}$, ρ_s, μ_s, u_s, and $\dot{m}_{\text{rel},s}$ are segment path length, porosity, representative particle diameter, tributary porous area, local density, local viscosity, superficial porous velocity, and local released mass flow; release itself follows Eq. (15).

$\Delta p_{\text{sys,hb}}$ follows the original heating-network numbering through Eqs. (80) and (81): Eq. (80) defines each group branch-path pressure $\Delta p_{\text{path},g}$, and Eq. (81) defines the reported heating total as shared distribution plus the common/controlling branch-path requirement. $\Delta p_{\text{dist,common}}$ is common header/splitter-feed loss. g_{ctrl} is the group with the largest private-path pressure, so $\Delta p_{\text{path,ctrl}}$ is the controlling branch requirement used by the parallel-network closure.

$\Delta p_{\text{shutoff},x}$ is the datasheet shutoff pressure (maximum static pressure rise at zero flow; `shutoffPressure_inH2O` converted to Pa). $\dot{V}_{\text{rated},x}$ is the manufacturer rated flow (`ratedFlow_CFM`, converted to SI). $\Delta p_{\text{rated},x}$ is the static pressure rise paired with that same rated point (`ratedPressure_inH2O`, converted to Pa). These are two points on the same blower curve, not absolute pressure states. $\dot{V}_{\text{free},x}$ is explicit or inferred by Eq. (18.5C-1a2). $\Delta p_{\text{op},x}(d)$ is bounded operating pressure used for power. $P_{\text{shaft},x}(d)$ is shaft fluid power. η_x is total blower efficiency. $P_{\text{idle},x}$ is the blower baseline electrical draw at low flow while running (non-hydraulic parasitics), configured by `idlePower_W`;

it is nonnegative in code and only contributes when blower is enabled and solved flow is positive (Eq. (18.5C-1h)). $P_{\text{rated},x}$ is the motor rated electrical cap. $P_{x,\text{elec}}(d)$ is blower electrical input. $P_{h,\text{coil}}(d)$ is heater-coil electrical power from Eq. (31). $P_{\text{tot},h}(d)$ and $P_{\text{tot},c}(d)$ are total heating/cooling electrical powers. $\dot{V}_{\text{ach,ab}}(d)$ is achieved assist-blower flow used by the spot-cooler model. $\dot{m}(d) = \rho_{\text{air}}(T_\infty(d), p_\infty)\dot{V}_{\text{ach,ab}}(d)$ is spot-cooler moist-air mass flow, and $\dot{m}_{\text{dry}}(d) = \dot{m}(d)/(1 + w_{\text{in}}(d))$ is dry-air flow for psychrometrics. $\dot{Q}_{\text{spot,req}}(d)$, $\dot{Q}_{\text{spot,act}}(d)$, $\dot{Q}_{\text{spot,max}}(d)$, $c_p(d)$, $T_{\text{target,cfg}}$, χ_{derive}, $T_{\text{target,derived}}(d)$, $T_{\text{target}}(d)$, $T_{\text{supply,actual}}(d)$, $\dot{Q}_{\text{sensible}}(d)$, $P_{\text{in}}(d)$, and COP are spot-cooler load, capped load, available capacity, branch specific heat, configured target, derivation toggle, derived target, effective daily target, actual post-cap supply temperature, sensible cooling rate, cooler electrical input in COP definitions ($P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d)$), and coefficient of performance.

Step 18.5-C NOAA/config derivation note. Section 18.5 blower/equipment terms are not NOAA-derived. They come from configuration and physics re-solve outputs at each day ambient: fan-curve inputs ($\Delta p_{\text{shutoff},x}, \dot{V}_{\text{free},x}$), electrical parameters ($\eta_x, P_{\text{idle},x}, P_{\text{rated},x}$), and day-specific system pressure $\Delta p_{\text{sys},x}(\dot{V})$ from the thermal-fluid model. That system-pressure term uses the original "Friction and Pressure Losses" equations: release and porous closure (15), (62), (60), cooling pressure totals (56), (18.5C-1b0), (76), and heating branch/total pressure equations (80), (81).

Step 18.5-D: Duty fractions and applied net bed load.

$$D_h^*(d) = \frac{Q_{\text{load},h}(d)}{Q_{h,\text{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\text{load},c}(d)}{Q_{c,\text{cap}}(d)}. \tag{18.5D-1}$$

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1). \tag{18.5D-2}$$

$$Q_{\text{bed,total,applied}}(d) = \begin{cases} D_h(d)\, Q_{h,\text{cap}}(d), & \text{heating day,} \\ -D_c(d)\, Q_{c,\text{cap}}(d), & \text{cooling day,} \\ 0, & \text{idle day.} \end{cases} \tag{18.5D-3}$$

$$Q_{\text{unmet},h}(d) = \max(Q_{\text{load},h}(d) - Q_{h,\text{cap}}(d), 0)\,, \tag{18.5D-4}$$

$$Q_{\text{unmet},c}(d) = \max(Q_{\text{load},c}(d) - Q_{c,\text{cap}}(d), 0)\,. \tag{18.5D-5}$$

18.6 Daily Bed Temperature, Electricity, Cost, and Water (Docx Notation)

Step 18.6-A: Split net bed load between bottom/top nodes.

$$Q_{\text{bed},b}(d) = f_b\, Q_{\text{bed,total,applied}}(d), \qquad Q_{\text{bed},t}(d) = (1 - f_b)\, Q_{\text{bed,total,applied}}(d). \tag{18.6A-1}$$

Step 18.6-B: Solve daily two-node bed temperatures. Using the report's Eq. (53)-style matrix, solved day-by-day:

$$\begin{bmatrix} (UA)_{b,\infty} + (UA)_{12} & -(UA)_{12} \\ -(UA)_{12} & (UA)_{t,\infty} + (UA)_{12} \end{bmatrix} \begin{bmatrix} T_{\text{bed},b}(d) \\ T_{\text{bed},t}(d) \end{bmatrix} = \begin{bmatrix} Q_{\text{bed},b}(d) + (UA)_{b,\infty} T_\infty(d) \\ Q_{\text{bed},t}(d) + (UA)_{t,\infty} T_\infty(d) \end{bmatrix}. \tag{53}$$

Step 18.6-C: Daily electricity and cost.

$$E_h(d) = \frac{24}{1000}\, P_{\text{tot},h}(d)\, D_h(d), \qquad E_c(d) = \frac{24}{1000}\, P_{\text{tot},c}(d)\, D_c(d), \tag{18.6C-1}$$

$$E_{\text{day}}(d) = E_h(d) + E_c(d), \tag{18.6C-2}$$

$$C(d) = c_{\text{elec}}\, E_{\text{day}}(d). \tag{18.6C-3}$$

Step 18.6-D: Daily water replacement in report notation. Using the report's evaporation framework:

$$\dot{m}_{\mathrm{diff},i}(d) = h_{m,i}(d)A_{\mathrm{seg},i}(d)\Delta\rho_{v,i}(d), \quad \dot{m}_{\mathrm{carr},i}(d) = \dot{V}_{\mathrm{rel},i}(d)\Delta\rho_{v,i}(d), \quad \dot{m}_{\mathrm{heat},i}(d) = \frac{\dot{m}_{\mathrm{rel},i}(d)C_P(T_{\mathrm{air},i}(d) - T_{\mathrm{bed,ref}}(}{h_{fg,i}(d)} \tag{107}$$

$$\dot{m}_{\mathrm{evap},i}(d) = \min(\dot{m}_{\mathrm{diff},i}(d),\, \dot{m}_{\mathrm{carr},i}(d),\, \dot{m}_{\mathrm{heat},i}(d))\,, \tag{108}$$

$$m_{\mathrm{water,replace}}(d) = \Delta t_d \sum_{i=1}^{N_{\mathrm{aer}}} \dot{m}_{\mathrm{evap},i}(d), \qquad \Delta t_d = 24 \text{ h}. \tag{18.6D-1}$$

If the run uses the simplified duty-scaled water approximation, keep:

$$m_{\mathrm{water,replace}}(d) = \dot{m}_{\mathrm{water},24h}\, D(d), \tag{18.6D-2}$$

with $D(d)$ equal to the active branch duty ($D_h(d)$ on heating days, $D_c(d)$ on cooling days, else 0).

Annual Sums (unchanged logic)

$$E_{h,\mathrm{yr}} = \sum_{d=1}^{365} E_h(d), \quad E_{c,\mathrm{yr}} = \sum_{d=1}^{365} E_c(d), \quad C_{\mathrm{yr}} = \sum_{d=1}^{365} C(d), \quad m_{\mathrm{water,yr}} = \sum_{d=1}^{365} m_{\mathrm{water,replace}}(d). \tag{18.6E-1}$$